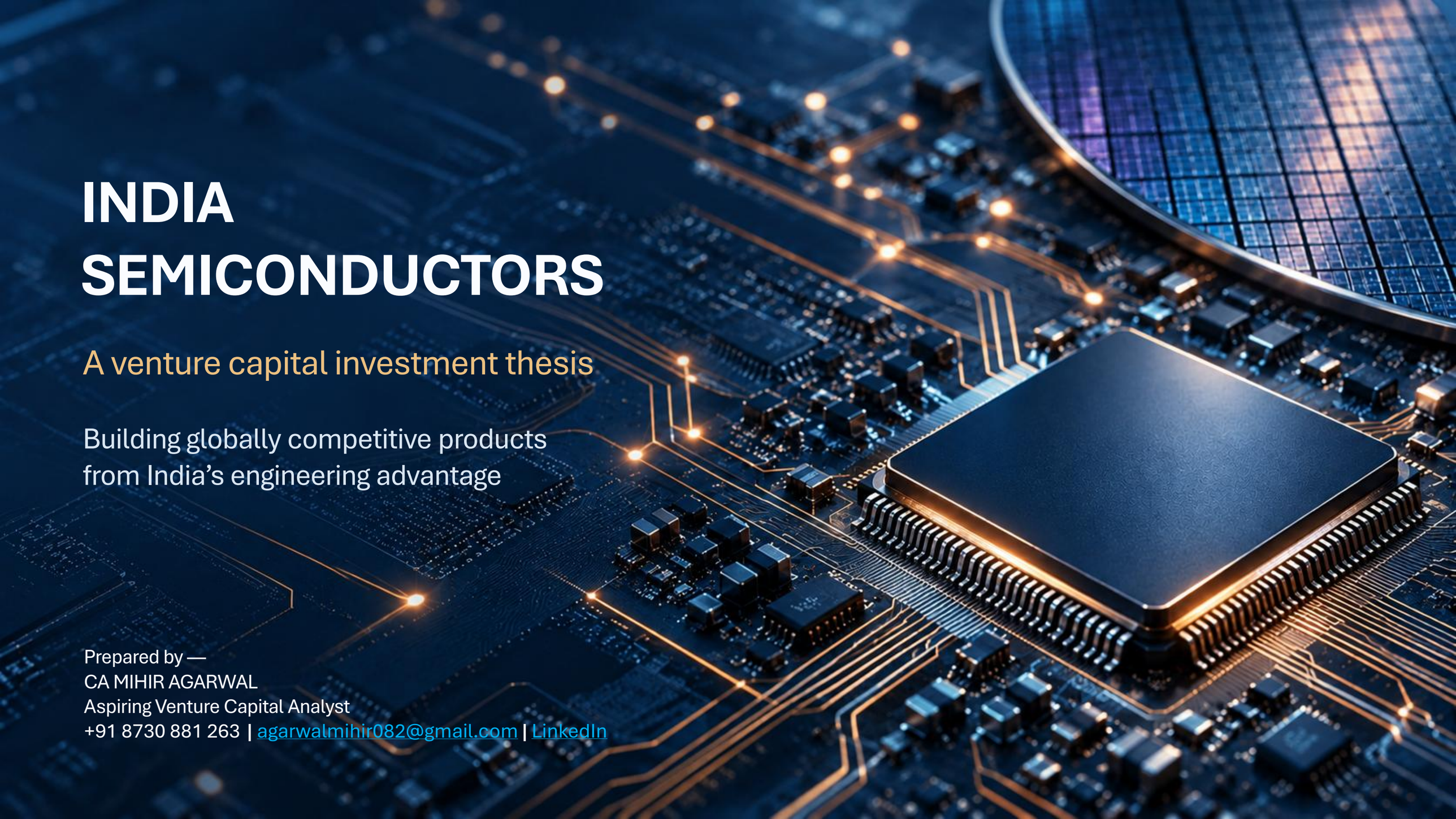


INDIA SEMICONDUCTORS

A venture capital investment thesis

Building globally competitive products
from India’s engineering advantage

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EXECUTIVE SUMMARY

India's chip opportunity is in owned technology, not in assembly alone

Indian chip demand is forecast to roughly double from $44bn (FY26) to $90bn (FY30), yet 90-95% of it is imported

Key problems in the sector

Import dependence: India imported about $30.3bn of IC and $6.2bn of devices in FY26.

Talent without ownership: about 20% of the world's chip designers are Indian, but most on customer-owned IP in global R&D centres.

Thin growth capital: Indian semiconductor startups raised ~$206m across 51 rounds since 2022, including $61.9m in H1 2026. Seven recent Series A rounds accounted for $73.7m, while funding beyond Series A remains relatively limited.

Imported inputs: wafers, tools, gases and chemicals for new fabs and packaging plants are largely imported.

Areas of focus for a VC fund

Power and analog chips for EVs, servers and industrial systems.

Secure, RISC-V and edge-AI chips for cameras, meters, telecom and defence.

Design tools and reusable IP with licence or royalty revenue.

Test, qualification and packaging services serving new plants in Gujarat and Assam.

Selected materials and tooling niches with qualified buyers.

Key impact and return potential

Import substitution: ~90–95% of India's semiconductor demand is currently import-dependent, leaving a substantial domestic supply gap as local demand scales.

Forex drain: ~$150bn spent on chips import from FY17 to FY25.

Strategic security: trusted silicon for defence, telecom and critical infrastructure.

Skilled jobs and IP: converts engineering services into owned products.

Exits: Global chipmakers and Indian strategics have demonstrated acquisition interest in semiconductor products, proprietary IP and design capabilities.

$90-155bn

Indian chip demand by FY30, from $44-64bn in FY26

~90-95%

of chips imported today

$206m

raised by chip startups since 2022

$8.8m

Average funding round in H1 2026, up from $3m in 2024

Thesis in one line: India offers a credible sourcing opportunity; returns require globally competitive products, owned IP and disciplined use of capital.

1.1 INDUSTRY OVERVIEW

India has the engineers. The next test is product ownership

India produced ₹13.11 lakh cr of electronics in FY26 but imported ~90-95% of the chips it uses

~20%
of global chip-design talent are Indian

12
manufacturing projects approved under the Semicon 1.0 (about ₹1.64 lakh cr)

₹13.11 lakh cr
electronics production in FY26, up from ₹1.9 lakh cr in FY15

~90-95%
of domestic chip demand met by imports

Dimension	Background	Current state (2026)	Investment implication
Global industry	Chips are made by design houses, foundries, packaging plants and tool suppliers; design captures about half of value with about 13% of capex	WSTS reports the semiconductor market reached $702bn in H1 2026; AI demand is the main growth engine, high power computing and advanced memory tech also adds	Design-led, capital-light models fit venture economics better than fabs
Indian base	Decades of engineering services and global R&D centres built a large design workforce; fab attempts stalled in the 1989	ISM launched Dec 2021; Micron, Tata and CG Power among other in Gujarat and Assam to get approval	India has deep chip-design talent, but limited experience in building IP, converting designs into products and securing global design wins
Startups	Fabless startups grew from a handful to 105+ recognised firms, however most remain pre-revenue	$206m raised in 51 rounds (2022 to H1 2026); average cheque up from $3.0m (2024) to $8.8m (H1 2026)	Seed and Series A active; growth capital pretty inactive
Bottlenecks	Imported wafers, equipment, gases and chemicals (over 90%); NITI calling for stronger sovereign design, advanced research and semiconductor R&D capabilities, currently only at ~0.6% of GDP	Power and water reliability and fab-ready technicians remain constraints	Test, qualification and selected supplier niches (chemicals, gases, spares) can grow with local plants

Key trends to watch

- AI infrastructure is pulling demand for power delivery, advanced packaging and memory.
- Open RISC-V architectures and chiplets lower entry barriers for design startups.
- ‘China+1’ supply-chain shifts are bringing assembly and test capacity to India.

What is new since 2021

- Semicon 1.0 launched with about $10bn; Semicon 2.0 adds about $13.4bn.
- Indigenous 180nm processors such as IRIS-LV and VIKRAM3201 demonstrated domestic design-to-fabrication capability.
- Startup funding rising: $61.9m in H1 2026 alone.

Why it matters to a VC

- Owning IP turns engineering fees into reusable revenue.
- Capital-light design fits venture economics; fabs need strategic capital.
- Customers and qualification, not talent, are the scarce resource. Startups need anchor customers, design wins and lengthy product qualification before revenues can scale.

1.2 INDUSTRY OVERVIEW: VALUE CHAIN

Value accrues to different owners at each stage

Design creates about 50% of chip value with about 13% of capex, while fabrication absorbs about 77% of capex

Layer	Revenue model	Global leaders	India today	Capture feasibility/ Capital requirement	VC relevance
Design tools (EDA) and reusable IP: the software and blocks used to design chips	Software licences, subscriptions, royalties	Synopsys, Cadence, Arm, Imagination	105+ startups; emerging domestic products; AI-assisted design startups	High/ low	Strong when technology is reusable i.e. saleable to many firms
Chip design and fabless products: designing and selling chips	Engineering fees, chip sales, royalties (high gross margin)	Nvidia, Qualcomm, Broadcom, AMD	Services established; products emerging; about 20% of designers	High/ low	Core focus: power, analog, edge AI, RISC-V
Wafer fabrication: the factory that makes chips	Per-wafer fees; profit depends on utilisation (break-even near 70% utilisation)	TSMC, Samsung, GlobalFoundries	First 28nm fab under construction targeting 50,000 wafer/month (Tata-PSMC, Dholera)	Low/ very high	Mostly strategic and project capital
Packaging and testing (OSAT/ ATMP): assembly, marking, testing, packing chips	Per-chip processing and test fees	Advantest, Amkor	Micron Sanand, Tata Assam, CG Power-Renesas, Kaynes	Medium/ medium	Specialist test tools, services for plants
Equipment and materials: tools, gases, chemicals, wafers	Tool sales, service contracts, consumables	ASML, Applied, Lam	Over 90% imported	Medium-low/ high	Selective niches with qualified buyers

Why the economics differ

- Design creates about 50% of value and absorbs about 53% of industry R&D, but only about 13% of capex; fabrication takes about 77% of capex.
- Fabless example: $6m design cost, $4.00 price and $2.20 unit cost give break-even at about 3.3m chips.
- Enabling suppliers serve fabs and packaging plants; electronics makers integrate chips into products and create the end demand.

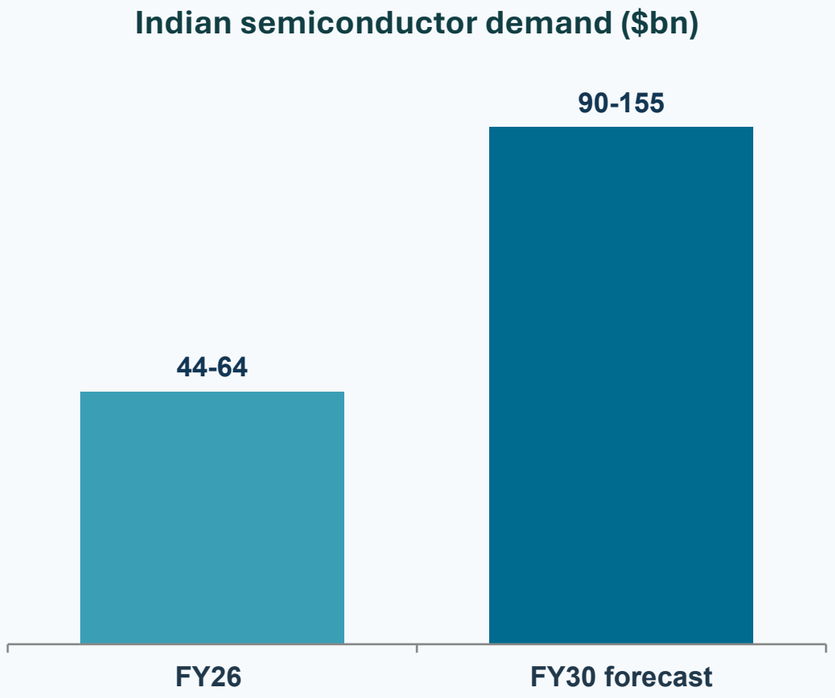
What India can realistically capture (next 10 years)

- Near term: chip design (IP), design tools (EDA), MPW, packaging and test services.
- Medium term: mature-node fabs (28 nm and above), power and compound semiconductors, photonics, RF/ 6G devices, chemical and gases.
- Long term: materials, sub-systems and equipment, leading-edge/ advanced nodes and memory fabs (not realistic venture targets).

2.1 MARKET SIZE AND GROWTH POTENTIAL

Domestic demand can roughly double by FY30

Sources define 'demand' differently, so this particular slide uses a range of current and forecasted economy



India's demand creates a customer base. Captured revenue still depends on the chip, the buyer and the qualification route.

~19.6% to 24.75%

implied annual growth over four years

$1.51tn

implied 2026 global sales from WSTS Q2 run-rate

Source	Base	Forecast
KPMG-IDTA/ NITI Aayog	$44bn (FY26)	$90bn (FY30); about $200bn (FY35)
Inc42 2026	$62bn (2026)	$155bn (2031), 9-10% global share
3one4 Capital 2026	$45bn (2021)	$110bn (2030)
EY 2026	$64bn (2026)	$200bn (2035)

Note: Forecasts differ by definition (chip sales versus chips inside electronics), base year and currency, so I have chosen to treat them as a range for this particular slide.

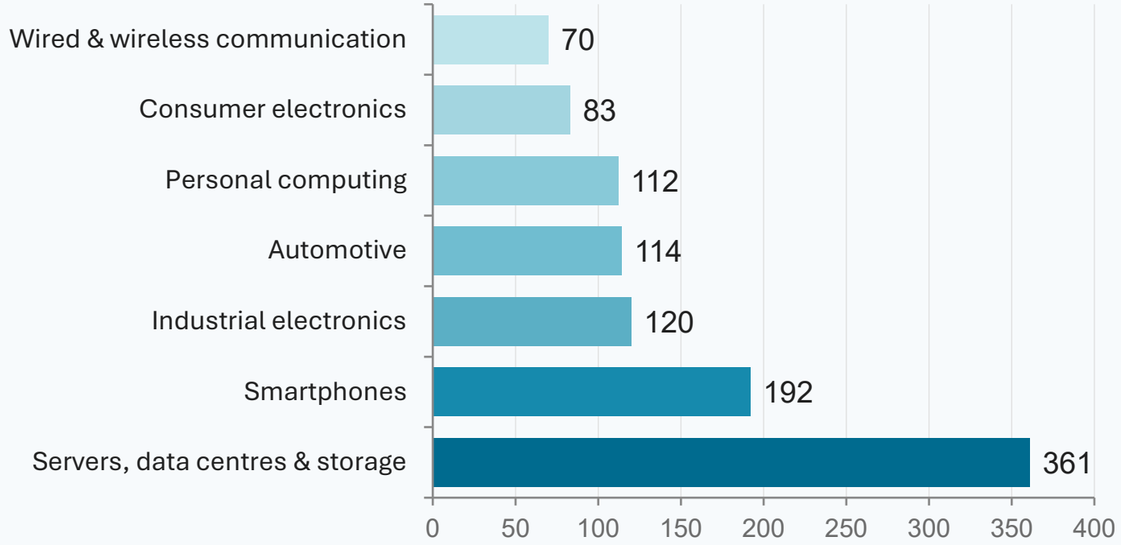
Reading the numbers

- WSTS forecasts global semiconductor sales to grow another ~27% in 2027 to ~$1.9tn, following exceptionally strong AI-led growth in 2026, with memory and logic the largest contributors.
- India's share is about 5-7% of global demand today, NITI Aayog aims for 10-13% by 2035.
- India is growing faster than the global market: the NITI trajectory implies India's demand could roughly double in four years, increasing its importance as an end-market for global and domestic chip suppliers.

2.2 MARKET SIZE AND GROWTH POTENTIAL: SEGMENT-WISE SIZING

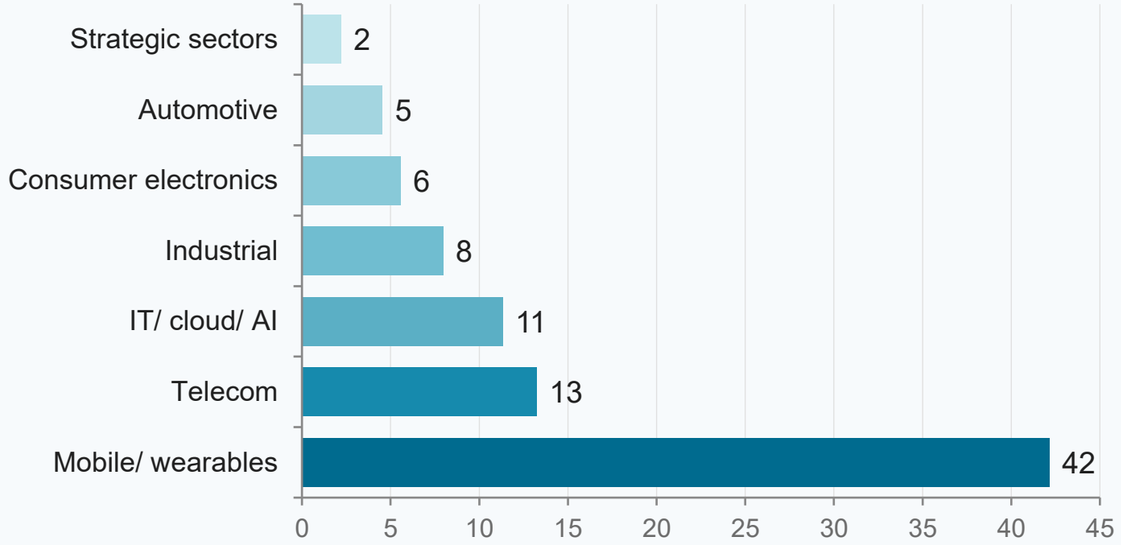
India's chip demand is forecasted at $90bn in FY30 (Niti): mobiles/ wearables almost half at $42bn

Global chip demand by segment, 2030 ($bn) – *ASML report*



Segment	2030 ($bn)	Share
Servers, data centre & storage	361	34.32%
Smartphones	192	18.25%
Industrial electronics	120	11.41%
Automotive	114	10.84%
Personal computing	112	10.65%
Consumer electronics	83	7.89%
Wired & wireless communication	70	6.65%

India chip demand by segment, FY30 ($bn) – *IDTA report*



Segment	FY30 ($bn)	Share
Mobile/ wearables	42.13	48.50%
Telecom	13.22	15.22%
IT/ cloud/ AI	11.31	13.02%
Industrial	7.95	9.15%
Consumer electronics	5.55	6.39%
Automotive	4.52	5.20%
Strategic sectors	2.19	2.52%

3.1 DRIVERS OF GROWTH

India produced phones worth ₹6.27 lakh cr in FY26, out of which exports were ₹2.60 lakh cr, EV and AI also driving semiconductor demand

Growth driver	Factors to consider	What the buyer needs	What India can capture (my assessment)
EVs, industrial equipment and renewables	EVs carry 1,000-3,000+ chips; automotive demand $4.52bn by FY30	Reliable power conversion and sensing	Analog and power chips, test expertise
AI servers and data centres	AI infrastructure needs about $5.2tn by 2030; power is a bottleneck; India's data-centre pledges over $200bn	Lower power use and faster data movement	Power delivery, connectivity IP, edge chips
Smartphones, cameras and IoT (consumer behaviour)	India assembles about 20-25% of iPhones; 150m+ phones shipped in a year	Low-cost, secure, locally supported chips	Edge AI and RISC-V SoCs
Defence, space and telecom (sovereign demand)	Trusted-source and preferential procurement rules; 5G roll-out	Trusted, custom components, long supply life	RF chips, sensors, specialised processors
New fabs and packaging plants	12 approved projects, about ₹1.64 lakh cr	Local service and qualified inputs	Inspection, test, tools, consumables

Emerging technologies

- Niti identifies chiplets and advanced packaging, RISC-V open architectures, GaN/ SiC power devices, RF/6G chip devices, photonics, edge AI, AI-assisted chip design.
- Only about 8% of India's tracked patents sit outside accelerators, so packaging and chiplets are open space.
- IIT madras is the only Indian institute to top in the list of patent assignees in India.

Consumer and enterprise behaviour

- Anchor domestic OEMs can accelerate adoption through co-development and early design wins; NITI recommends structured startup–OEM co-development frameworks.
- Buyers still compare reliability, total system cost and delivery guarantees; 'made in India' alone does not lower switching risk.

Three phases

- Before 2021: Global design/services base built engineering depth, limited domestic product/IP ownership.
- 2021-2026: Policy incentives, design support and manufacturing/packaging capacity establish the ecosystem.
- 2027-2030: Qualified suppliers, design wins and repeat sales to determine actual value capture.

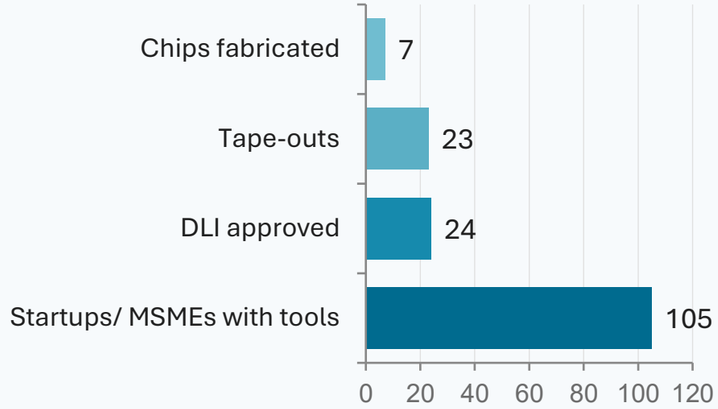
3.2 DRIVERS OF GROWTH: REGULATION AND POLICY

Semicon 2.0 (₹1.27 lakh cr,) is 1.7 times Semicon 1.0 (₹76k cr) and adds six pillars of support

Programme outlays, project incentives and talent initiatives serve different purposes

Support layer	Scale/ mechanism	What changes for a startup
Semicon program	• **1.0**: ₹76k cr outlay and establishment of ISM • **2.0**: about ₹1.27 lakh cr outlay	1.0 gave the initial boost, 2.0 is set to carry the legacy forward
Semicon 2.0 (Six pillars)	• **Design**: *(detailed below)* • **Materials/ equipment**: 30% of capex and 2-10% PLI on domestically sourced BoM for five years from 2028-29 • **Fabs**: 40% of capex on silicon fab; 35% for compound semiconductors, sensors, etc. • **ATMP/ OSAT**: 35% of capex for advanced packaging and 25% for legacy packaging • **R&D**: Funding upto 75% of project cost for advanced R&D • **Talent**: Funding upto 75% of project cost for educating and workforce-developing initiatives upto 6 years	• Reduces net equity burden for eligible projects • Provides certain tools and equipment for designing
Design Linked Incentive (integral part of Semicon 2.0)	• **Strategic/ National importance:** Product specific financial support • **Design**: Tools/ prototype and milestone linked seed funding of 50% of project cost upto ₹15cr and co-investment beyond that or royalty/ co-investment • **Deployment**: Reimbursement of 9% of net sales upto ₹120cr/company for 5 years • **Semicon 1.0**: Approved 24 design projects and provided EDA tools to 105 startups/ MSMEs	Funds design and rewards commercial sales
C2S and academic infrastructure	320 institutions with EDA tool access, 68,000+ students trained.	Lowers access barriers to design tools, training and prototyping
Wider deep-tech support (RDI) , ECMS and others	₹1 lakh cr RDI scheme; ECMS raised to ₹40k cr; IndiaAI Mission	Long-term low/ nil-interest loans, equity, Deep-Tech FoF participation and tax incentives for 3 years
State incentives/ electronics PLI	Project-specific and adjacent support in Gujarat, Assam, Uttar Pradesh and others	Improves local economics and end-customer scale

Design support outcomes (count)



DLI support is associated with private follow-on capital

- 14 of 24 DLI-supported companies have raised venture funding (about $100.8m).
- 6 companies secured follow-on rounds worth $53.6m, suggesting some investors were willing to recommit after initial technical milestones.
- Four companies hold about 56% of that capital i.e. funding is concentrated.
- Government selection/ support can signal a positive diligence, but cannot be considered as conclusive investment validation.

4.1 KEY PLAYERS: PUBLIC COMPANIES

Listed Indian strategics have demonstrated acquisition appetite for semiconductor design/ IP acquisitions, while CG Power and Kaynes have commissioned major OSAT facilities at Sanand

These are ecosystem reference points, not stock recommendations or direct valuation comparables

Listed reference	Relevant exposure	What has happened	To be checked from finance angle
CG Power	Its semiconductor venture adds assembly and testing exposure to an established industrial group.	OSAT JV with Renesas and Stars of about ₹7.6k cr at Sanand; bought Renesas' RF unit	Plant utilisation, customer mix and capital still to deploy
Kaynes Technology	Electronics manufacturing provides the base for expansion into semiconductor assembly and testing	Electronics manufacturer; ₹3.3k cr Sanand OSAT approved Sep 2024; 6.3m chips a day capacity	Same as above
Cyient	Engineering services and semiconductor investment activity provide a services-to-product transition case	Has a semiconductor subsidiary; bought stake in Azimuth AI for $7.25m; majority investment in Kinetic (about $85m, Apr 2026)	Whether engineering capability can convert into proprietary IP, product revenue and higher-quality margins
MosChip	Chip design and embedded engineering services offer exposure to the engineering portion of the value chain	Semiconductor design services, IP and turnkey ASIC; transitioning from pure-play design services toward turnkey ASIC solutions	Repeat IP revenue versus turnkey project-based engineering fees
HCL Tech (Group)	Semiconductor engineering services support global customers across design and related technology work	Semiconductor and equipment engineering services; bought Sankalp Semi (about $25m, 2019); HCL-Foxconn display-driver JV at Jewar	Revenue per engineer, contract duration and reusable assets
Tejas Networks; Infosys; L&T	Acquirers of chip businesses	Tejas-Saankhya (about $58m, 2022); Infosys-Insemi (about $33m, 2024); L&T-SiliConch (about $22m, 2024)	Evidence that strategic M&A can provide an exit route for Indian design/ IP startups

Why they matter for a VC

- Listed strategics demonstrate credible domestic M&A exit routes for semiconductor design and IP startups.
- OSAT players reveal whether utilisation, yield and customer qualification can turn subsidised capacity into viable economics.
- Design-services players show whether engineering talent can be converted into reusable IP, products and recurring revenue.
- These companies can be used as business-model reference points—not direct valuation comparables for venture-backed startups.

4.2 GLOBAL PLAYERS: COMPETITIVE LANDSCAPE & M&A ACTIVITY

Global leaders dominate every layer; Indian chip exits so far range vastly, mostly to strategic buyers

Global incumbents already have trusted products, broad catalogues and established support channels

Arena	Global reference competitors	Indian players and entry wedge	Evidence that would support a moat	VC fit
Processor IP, RISC-V and EDA tools	Arm, Imagination, Synopsys, Cadence	InCore, Mindgrove; Maieutic	Reusable design wins and repeat licence revenue	★★★★☆
Analog, power, RF and compound fabs	TI, ADI, Infineon, NXP, Renesas	C2i, AGNIT, Fermionic, Vervesemi	Measured system benefit and qualification	★★★★☆
AI and edge compute	NVIDIA, AMD, Qualcomm, Broadcom	Netrasemi, BigEndian, Agrani Labs	Software compatibility and customer-level economics	★★★☆☆
Packaging, equipment and materials	Amkor, Teradyne, Advantest	Tata, Micron, Kaynes	Process know-how and recurring supplier approvals	★★★☆☆
Wafer fabrication	TSMC, UMC, GlobalFoundries	Tata-PSMC (Dholera); mature-node and strategic capacity	Yield, utilisation and funded long-term operations	★☆☆☆☆

Indian semiconductor startups M&A exits

- **Steradian-Renesas** ~₹417cr
- **Saankhya-Tejas (~64.40%)** ~₹284cr
- **C2i-Infineon** Undisclosed
- **Kinara-NXP** ~$307m
- **Sankalp-HCL** ₹180cr
- **SiliConch-L&T** ~₹178cr
- **InSemi-Infosys** Upto ₹280cr
- **Cerium-Tech M (EV)** Upto ~₹245cr

What the M&A precedents tell us

- **Two exit pathways**: Global companies acquire product technology and proprietary IP, while Indian IT/ engineering groups acquire semiconductor design capabilities.
- **Product-IP exits can command substantial values**: Steradian and Kinara demonstrate strategic demand for differentiated chips, although Kinara is a US-incorporated company and not directly comparable to India-headquartered startups.
- **Exit values are not directly comparable**: Disclosed consideration includes outright acquisitions, staged purchases, earn-outs and contingent payments. A sector-wide average exit valuation would be misleading.

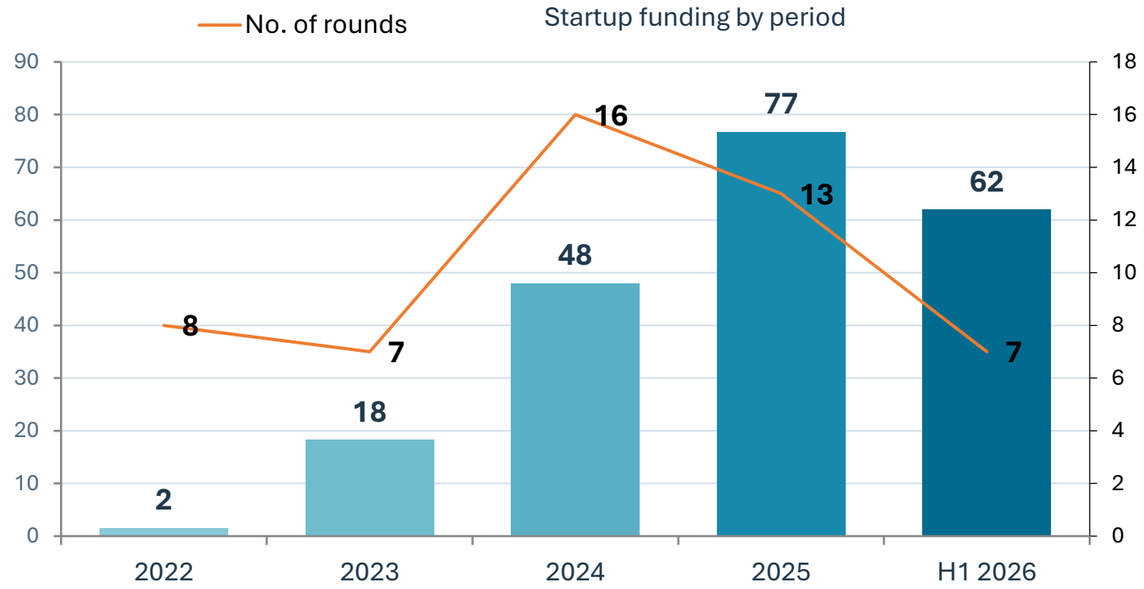
What makes a startup acquisition worthy?

- **Proprietary tech**: Patents, know-how or differentiated chip structure that is difficult to replicate.
- **Strategic portfolio fit**: Technology that fills a product gap or accelerates the acquirer's roadmap.
- **Commercial validation**: Qualified design wins, repeat orders or credible customer adoption.
- **Scalable economics**: Attractive gross margins, manageable capital needs and a path to volume production

5.1 STATE OF INDIAN SEMICONDUCTOR STARTUPS

Venture funding and ecosystem development

Capital is rising and policy has lowered the cost of first chip



$206m

raised across 51 rounds from 2022 to H1 2026

$3.0m to $8.8m

average round size from 2024 to H1 2026

$61.9m

H1 2026 funding, already 81% of full-year 2025

Where the sector stands

- Funding is rising, led by seven Series A rounds aggregating to $73.7m in edge AI, power and RISC-V.
- Approved policy reimbursement pays for the first chip: DLI awards fund design and tape-out, so an award is a screening signal.
- Founders come from university labs, repeat founders with past exits, and industry veterans.
- Corporate investors and strategic acquirers are increasingly participating in India's semiconductor ecosystem.

What is still missing

- Tape-outs and prototypes have increased, but sustained production orders and customer-level unit economics remain insufficiently disclosed.
- Capital is concentrated: four companies (C2i, NetraSemi, Morphing Machines and Mindgrove) hold about 56% of DLI-linked venture funding.
- Larger rounds do not eliminate the need for additional funding for qualification, packaging, inventory and customer acquisition.
- Strategic M&A provides precedents, but India-headquartered semiconductor product exits remain limited and disclosed valuations are not directly comparable.

5.2 WHERE INDIA CAN WIN

India's strongest VC opportunities may lie in chip design, reusable IP and chip infrastructure

Winning in this race would mean a product that sells again and again, not a single custom project

1 Design tools and verification

Design cycles generally run 1-3 years; AI-assisted tools can shorten them.

Buyer and model: Semiconductor design teams; software subscriptions or enterprise licences.

Capital intensity: Lower

2 Reusable chip IP

Customizable processor and interface IP can reduce development time and enable repeat licensing.

Buyer and model: Fabless chip designers and SoC developers; IP licensing, integration fees.

Capital intensity: Lower to medium

3 Power and analog chips

EV, industrial and data centre applications need efficient power management and precise sensing.

Buyer and model: OEMs, power-supply manufacturers and Tier-1 suppliers; chip sales.

Capital intensity: Medium

4 Secure RF and sensor chips

Defence, telecom and industrial systems need specialized RF and sensor ICs; over 90% of inputs are imported.

Buyer and model: Defence and telecom; chip sales, custom designs and possible licensing.

Capital intensity: Medium

5 Test and reliability

Expanding OSAT capacity creates potential demand for test engineering, inspection and reliability services.

Buyer and model: Chip firms and OSATs; testing and packaging fees

Capital intensity: Medium

6 Fab materials and parts

New fabs create opportunities for domestic suppliers of specialty materials, precision components and consumables.

Buyer and model: Fabs/ OSATs and tool makers; repeat supply

Capital intensity: Higher

Where disruption is plausible

- AI-assisted EDA can reduce repetitive design, verification and optimization work; actual time savings depend on chip complexity.
- India-specific chips for smart meters, cameras and e-mobility/ EVs.
- Fab and OSAT support tooling and metrology as plants ramp in Gujarat and Assam.

Harder for conventional VC

- Memory and leading-edge logic, where capital and scale dominate.
- One-off custom projects with no second buyer.

Investment filter: Measurable customer value, defensible technology, repeatable sales and a credible path to multiple buyers

6 HOW I WOULD EVALUATE STARTUPS

I screen through five gates; a sixth set of signals strengthens conviction

Evidence must be stage-appropriate. Each cheque funds the next measurable, risk-reducing milestone

1 Real buyer problem

Named buyer, measurable pain, credible switching benefit and purchase intent.

RED FLAG *Only TAM projections; no reachable buyer.*

2 Defensible IP & team

Clear commercial IP rights, differentiated design and experienced technical team.

RED FLAG *Unresolved IP rights or a missing core technical capability.*

3 Technical/ silicon proof

Stage-appropriate validation, foundry route and funded design-iteration plan.

RED FLAG *Unverified performance or no feasible fab path.*

4 Commercial proof

Customer evaluations advancing to design wins, qualification and paid production.

RED FLAG *Pilot activity with no route to orders.*

5 Funded path

Costed next milestone, contingency runway and realistic follow-on financing.

RED FLAG *Success relies on uncommitted grants or rescue rounds.*

6 Good to have

POSITIVE SIGNALS

- Repeat founder, ex-professor or ex-global chip industry experience
- Approved government support
- Strategic OEM/ foundry partners

IMPORTANT *Positive signals are not substitutes for passing the five gates.*

DISCLAIMER

- *This investment thesis has been prepared solely for educational, research and professional development purposes and does not constitute investment advice or a solicitation to invest.*
- *The analysis is based primarily on publicly available information, industry reports, company disclosures, proprietary data and other secondary sources. The accuracy and completeness of such information have not been independently verified in all cases.*
- *Market projections, valuations, and other forward-looking estimates involve assumptions and are subject to uncertainty. Actual outcomes may differ materially.*
- *The analysis reflects information available up to 30 September 2026, unless otherwise stated. Subsequent developments may affect the findings and conclusions.*